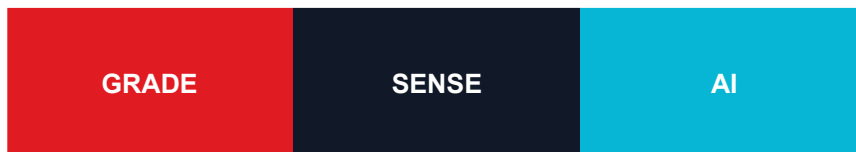


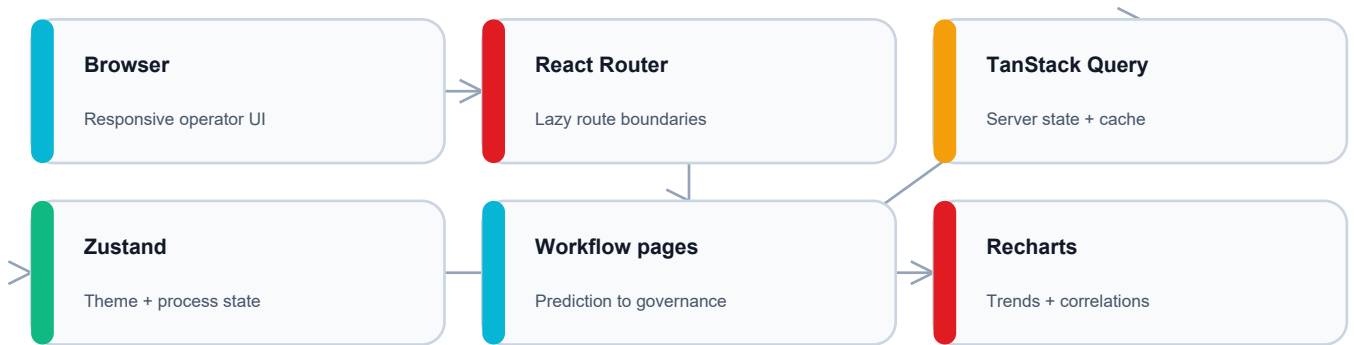
# GradeSense System Architecture

Seven professional architecture views across UI, services, ML, data, and inference

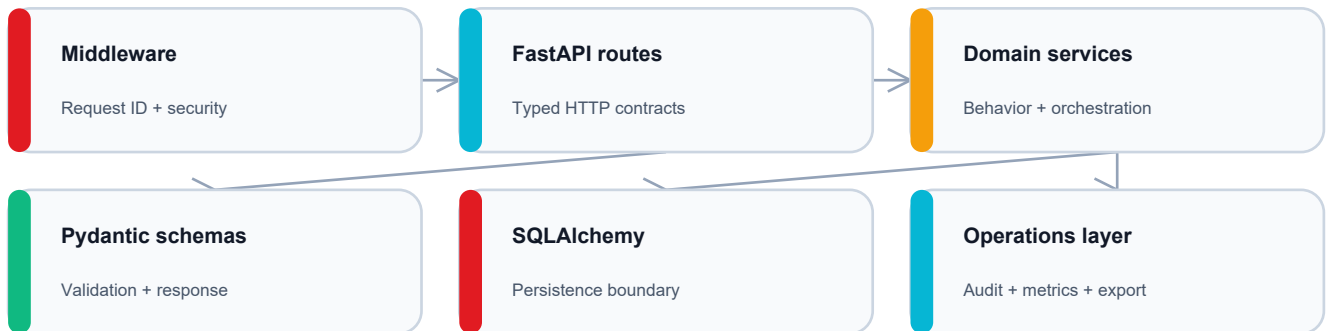


**Explainable predictions, constraint-aware recommendations, and real-time operations for safer, faster paper grade transitions.**

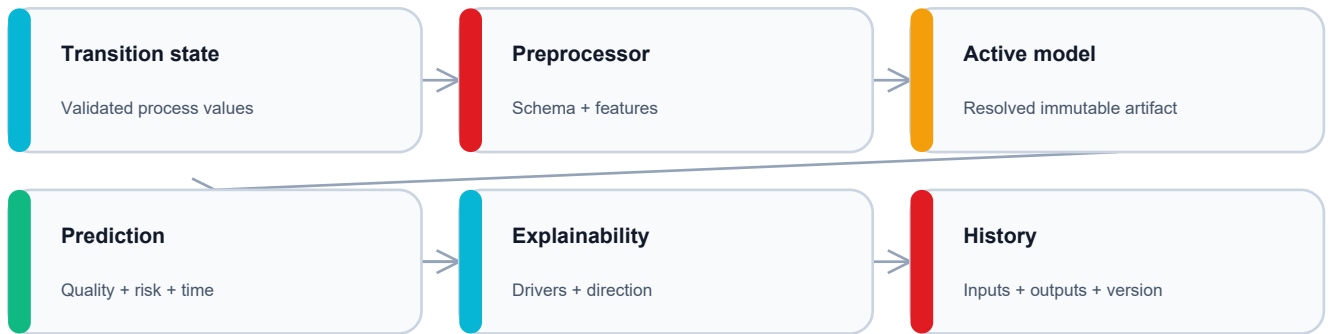
Prepared from the validated GradeSense v0.2.0 source repository and live application.



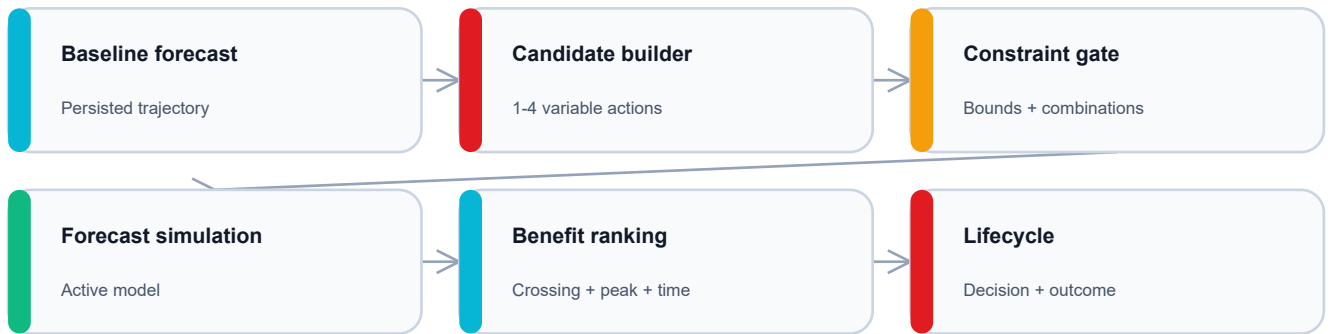
**Presentation, routing, server state, local state, and visualization are separated so each workflow stays testable.**



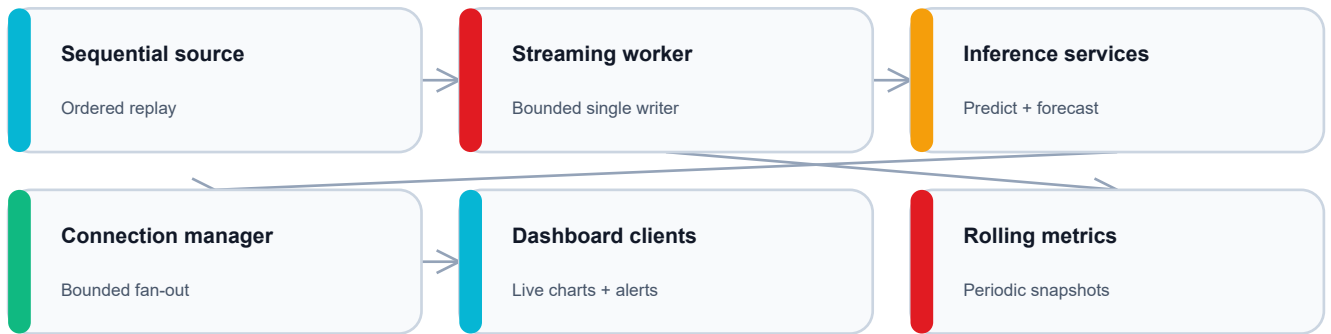
**Routes own transport semantics; services own behavior; schemas and ORM models define explicit contracts.**



**Inference is versioned, reproducible, explainable, and persisted with its input context.**



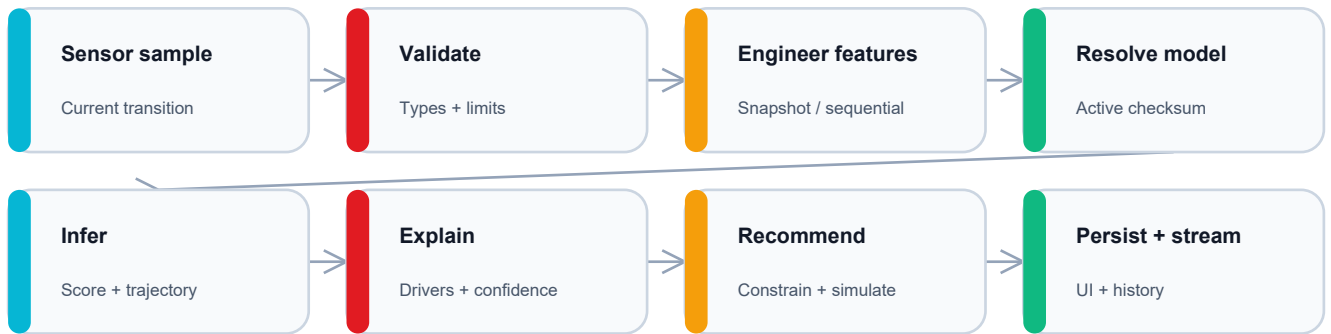
**No action reaches an operator until it passes constraints and improves a model-simulated baseline.**



**A bounded worker prevents unbounded memory growth and broadcasts typed events to connected clients.**



**Operational outcomes remain linked to the predictions and recommendations that produced them.**



The same governed flow connects a raw process observation to an explainable, advisory action.